TradePilot Root-Cause Analysis II

Trade-level forensics – the real culprit is over-trading and late entry, not the safeguards

Project	TradePilot (paper-trading, NSE intraday)
Report	Root-Cause Analysis II – forensic deep-dive (companion to 2026-06-24)
Version	`v2.0.0`
Status	Complete
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Method	3 parallel read-only investigations: trade-level forensics, FLAT_EXIT A/B, churn-cost reconciliation

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1. The Reframe

The first report suspected the bolted-on safeguards – especially FLAT_FORCE_EXIT – were the down-day bleed. Trade-level forensics **refute that.** Three independent investigations now converge on a different, simpler cause:

The engine trades too often and enters too late. A positive gross edge is consumed entirely by transaction cost and missed-by-late-entry profit.

- FLAT_FORCE_EXIT is **innocent** – on the two worst v5-specific days it actually protected capital.
- The real bleed is **entry timing**: on the same names, same direction, v5 enters 30–60 minutes later than the simple engine and captures far less.
- **Transaction cost flips a winning book into a losing one**: gross +Rs 4,691, cost –Rs 7,153, net –Rs 2,462 over 8 days. Excess churn alone was 205% of net P&L.

2. Evidence Stream A – Forensic Audit (06-17 and 06-22)

On both days the frozen original (v5_classic) made money while live v5 lost, on the same signals.

DAY	V5 GROSS	V5_CLASSIC GROSS	GAP
06-17	-766.71	+2,088.32	-2,855.03
06-22	-1,050.47	+273.99	-1,324.46

Attribution of the gap (reconciles exactly to the totals):

SUSPECT	06-17 VERDICT	06-22 VERDICT	CONCLUSION
FLAT_FORCE_EXIT	helped +30 vs classic	helped +299 vs classic	**Innocent – protected capital**
SHORT_BLOCK	saved +190 (suppressed losers)	inactive	**Innocent / helpful**
Wide 2.25% gap-stop	never fired	-764 (HUDCO, GVT&D)	**Minor real cost, one day only**
Late entry (same name/side)	**-1,969 of -2,855**	**-1,092 of -1,324**	**The dominant culprit**

The killer is the bottom row. v5 entered later than v5_classic on 16 of 35 shared positions (06-17) and 21 of 38 (06-22) – late enough that winners classic rode to TARGET (COFORGE, TMPV, DIXON, TATACAP) were booked by v5 as STOPLOSS, SIGNAL_FLIP, or TIME_EXIT instead. The 30-minute rescore-and-redeploy loop fragments and delays entries.

3. Evidence Stream B – FLAT_EXIT A/B (v5_apr, 9 days)

v5_apr runs identical v5 code with FLAT_EXIT disabled (and winner re-arm raised 3 to 6).

BASIS	V5	V5_APR	DIFFERENCE
Net P&L (9 days)	-341	+967	+1,308
Gross P&L (9 days)	+7,737	+7,817	+80

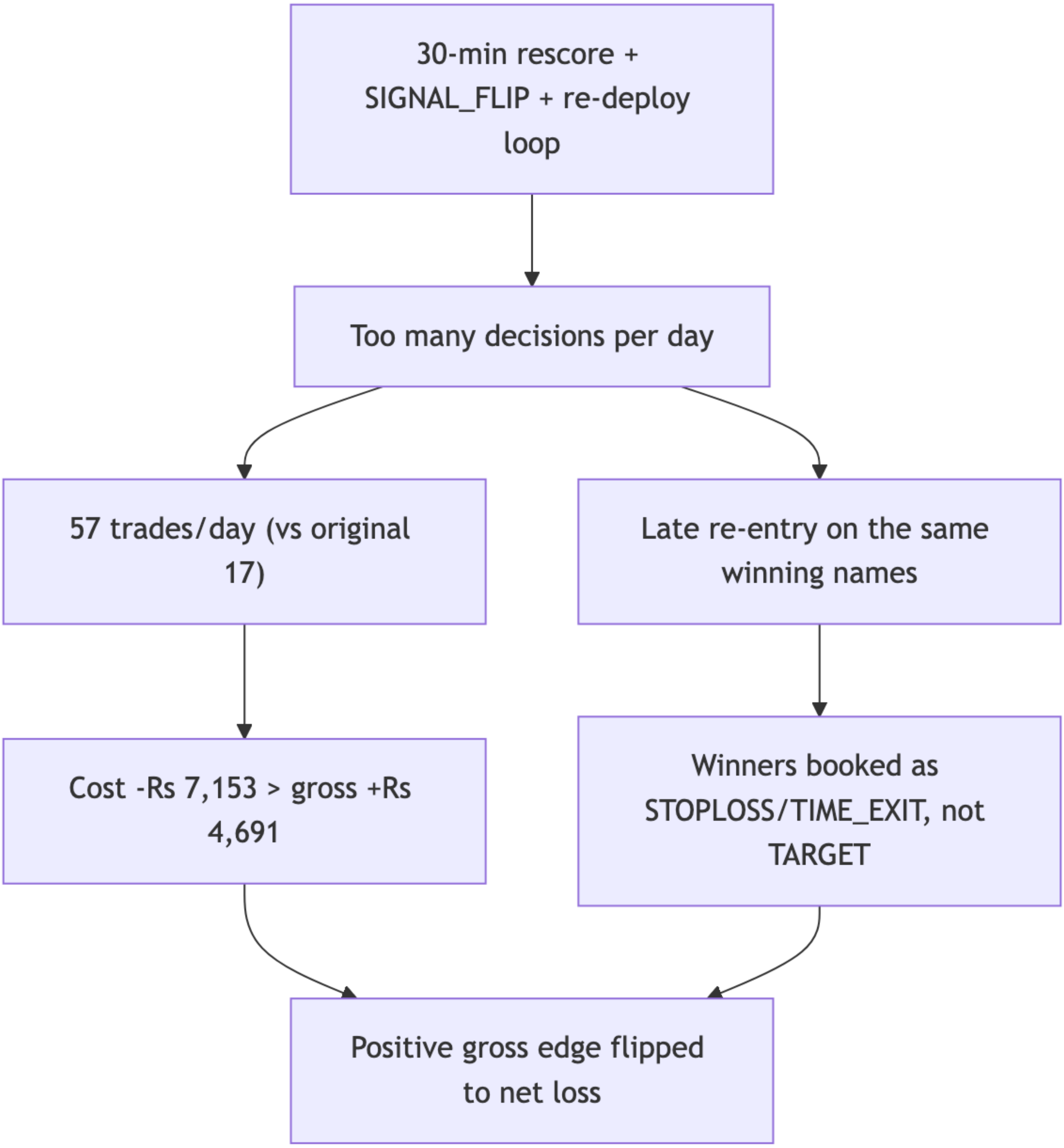
Two conclusions: (1) disabling FLAT_EXIT helps net and wins 6 of 9 days; (2) on a gross basis the engines are a coin-flip – **the entire +1,308 edge is cost/efficiency, not raw alpha.** Confound stated honestly: winner-rearm also changed, so this is a directional signal to run a clean single-variable test, not a settled verdict. n=9 is too small for significance.

4. Evidence Stream C – Churn-Cost (8 days, reconciled to the rupee)

METRIC	VALUE
Gross P&L	+4,691
Round-trip cost paid (12 bps x 454 trades)	-7,153
Net P&L	-2,462
Excess cost vs original 17-trades/day cadence	-5,046 (205% of net)
Counterfactual net at 17 trades/day	+2,584

Actual cadence is ~57 trades/day, not the 45 estimated earlier. The cost model is verified exactly – recomputing all 454 trades reproduces the reported Rs 7,153 to the rupee. A separate discovery: average position notional is **Rs 13,130, ~10x smaller** than the 15%-of-pool rule implies – so capital is barely deployed (~Rs 260k working on Rs 10L) and sliced into tiny lots churned 57x/day.

5. Unified Root Cause



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The original engine entered early, held, and exited on a tight bracket – ~17 trades/day, catching moves at the start. The current engine's re-evaluation machinery generates constant churn: it re-scores, flips, fragments, and re-enters, which (a) delays entry past the profitable window and (b) racks up cost that exceeds the gross edge. The safeguards we suspected are not the problem; **over-activity is.**

6. Revised Fix Priorities

ID	ACTION	PRIORITY
RC-1	Resurrect the original simple engine as a live shadow; 2-week head-to-head	High
RC-5	Slow or disable the 30-min rescore/ SIGNAL_FLIP loop; enter early and hold to bracket	High
RC-3	Cut trade count / concentrate capital (fewer, larger positions)	High
RC-6	Clean single-variable A/B: FLAT_EXIT=0 with rearm kept at 3	Medium
RC-2	Gate shorts off in SIDEWAYS regime	Medium
RC-4	Tighten or remove the 2.25% gap-day stop	Low

FLAT_FORCE_EXIT is dropped from the priority list – the evidence clears it. The high-value levers are all about **doing less**: trade less often, enter once and hold, concentrate capital. That is precisely what the profitable original did.

7. Key Learnings

THEME	LEARNING	APPLICATION
Evidence over plausibility	The diff-obvious suspect (FLAT_FORCE_EXIT) was exonerated by rupee-level audit; it helped, not hurt	Always attribute losses trade-by-trade before fixing a suspected mechanism
The real bleed	Late entry on the same winning names cost -1,969 and -1,092 on the two audited days	Enter at signal, hold; do not let a rescore loop delay or fragment entries
Cost is the killer	Gross +4,691 but cost -7,153 = net -2,462; excess churn = 205% of net	Trade count is a direct cost; cut cadence from 57 toward 17/day
Under-deployed capital	Avg position Rs 13,130 vs Rs 135k implied; ~Rs 260k working on Rs 10L	Fewer, larger positions – concentration, not fragmentation
Edge is intact	On gross terms the edge still exists; the loss is pure execution drag	The strategy is not broken; the trading frequency is